

Exp1: 1: Monthly Sales Data

a) R code:

```
# Dataset
```

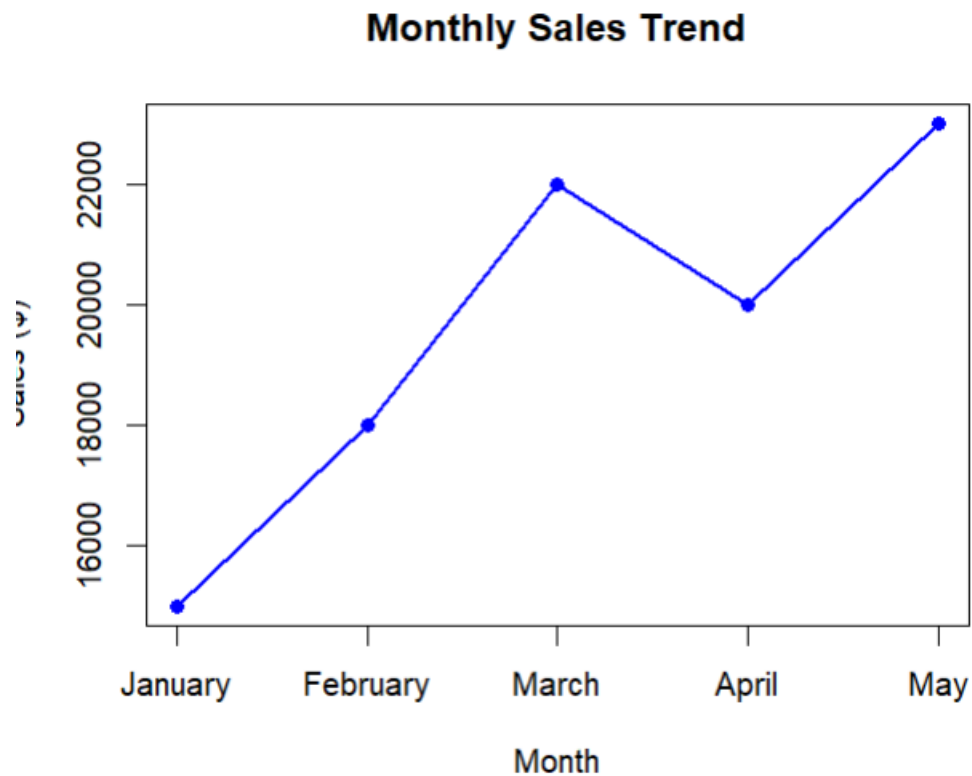
```
sales <- data.frame(  
  Month = c("January", "February", "March", "April", "May"),  
  Sales = c(15000, 18000, 22000, 20000, 23000)  
)
```

```
# Line Chart
```

```
plot(sales$Sales,  
  type = "o",  
  xaxt = "n",  
  xlab = "Month",  
  ylab = "Sales ($)",  
  main = "Monthly Sales Trend",  
  col = "blue",  
  lwd = 2,  
  pch = 16)
```

```
axis(1,  
  at = 1:nrow(sales),  
  labels = sales$Month)
```

output:



b) Generate a Bar Chart for Top-Selling Products

R code:

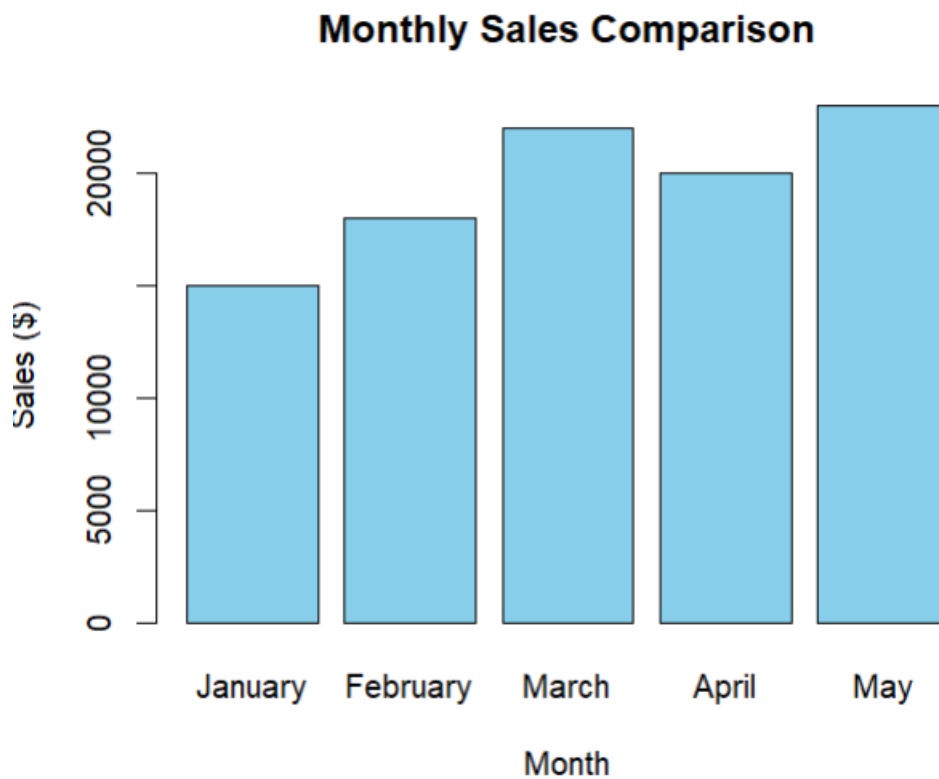
Dataset

```
sales <- data.frame(  
  Month = c("January", "February", "March", "April", "May"),  
  Sales = c(15000, 18000, 22000, 20000, 23000)  
)
```

Bar Chart

```
barplot(sales$Sales,  
  names.arg = sales$Month,  
  col = "skyblue",  
  xlab = "Month",  
  ylab = "Sales ($)",  
  main = "Monthly Sales Comparison")
```

output:



c) Scatter Plot Between Advertising Budget and Monthly Sales

R code:

Dataset

```
sales <- data.frame(  
  Month = c("January", "February", "March", "April", "May"),  
  Sales = c(15000, 18000, 22000, 20000, 23000),  
  Advertising = c(2000, 2500, 3000, 2800, 3500)  
)
```

Scatter Plot

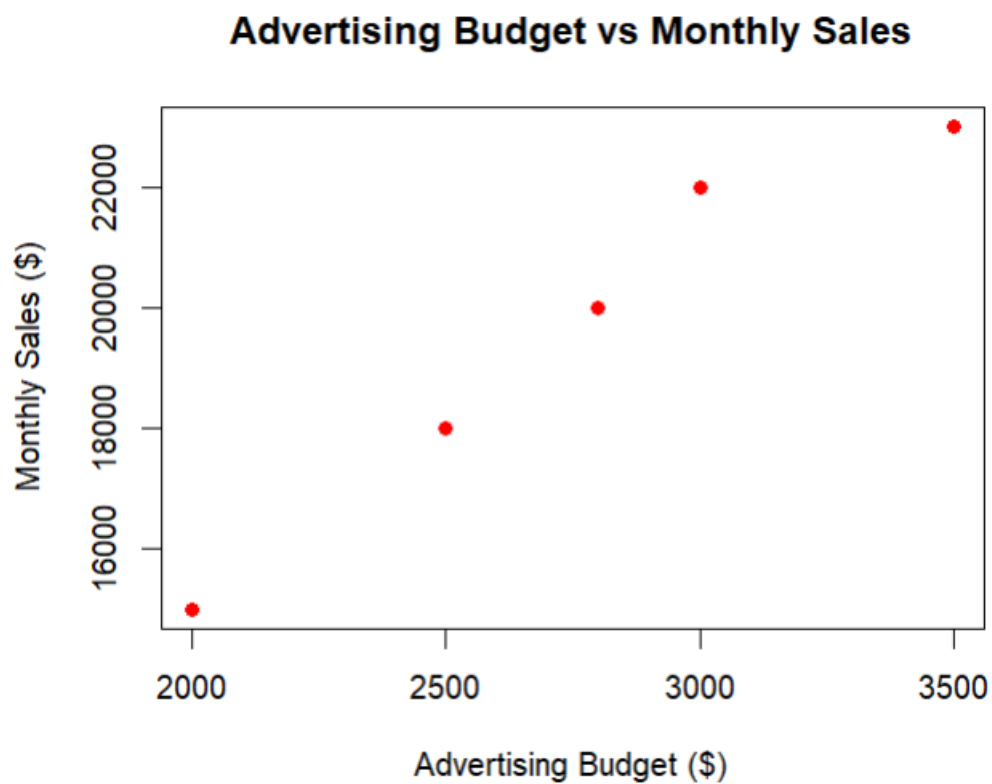
```
plot(sales$Advertising,  
     sales$Sales,  
     pch = 19,  
     col = "red",
```

```
xlab = "Advertising Budget ($)",  
ylab = "Monthly Sales ($)",  
main = "Advertising Budget vs Monthly Sales")
```

```
# Regression Line
```

```
abline(lm(Sales ~ Advertising, data = sales),  
       col = "blue",  
       lwd = 2)
```

output:



d) Interactive Dashboard (Using Shiny)

R code:

```
library(shiny)
```

```
sales <- data.frame(
```

```
  Month = c("January", "February", "March", "April", "May"),
```

```
Sales = c(15000, 18000, 22000, 20000, 23000)
)
```

```
ui <- fluidPage(

  titlePanel("Monthly Sales Dashboard"),

  sidebarLayout(

    sidebarPanel(

      selectInput("chart",
                  "Choose Chart:",
                  choices = c("Line Chart",
                              "Bar Chart"))

    ),

    mainPanel(

      plotOutput("salesPlot")

    )

  )

)
```

```
server <- function(input, output){

  output$salesPlot <- renderPlot({

    if(input$chart == "Line Chart"){

      plot(sales$Sales,
           type = "o",
           xaxt = "n",
           xlab = "Month",
           ylab = "Sales ($)",
           main = "Monthly Sales Trend",
           col = "blue",
           lwd = 2)

      axis(1,
           at = 1:nrow(sales),
           labels = sales$Month)

    }else{

      barplot(sales$Sales,
              names.arg = sales$Month,
              col = "orange",
              xlab = "Month",
              ylab = "Sales ($)",
              main = "Monthly Sales Comparison")
    }
  })
}
```

```

    }

  })

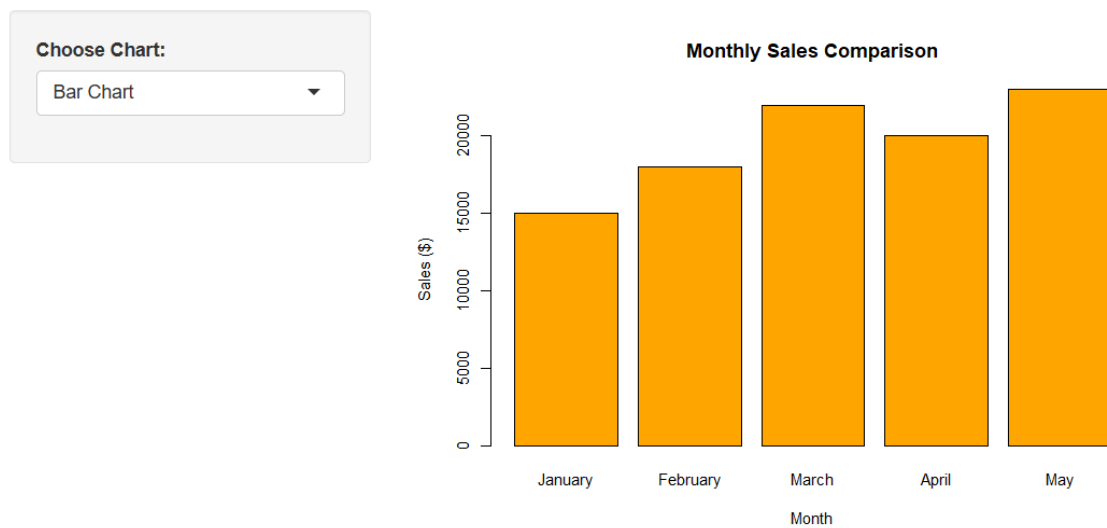
}

shinyApp(ui = ui, server = server)

output:

```

Monthly Sales Dashboard



Set 2: Customer Feedback Analysis

- Create a Histogram for Customer Ages

R code:

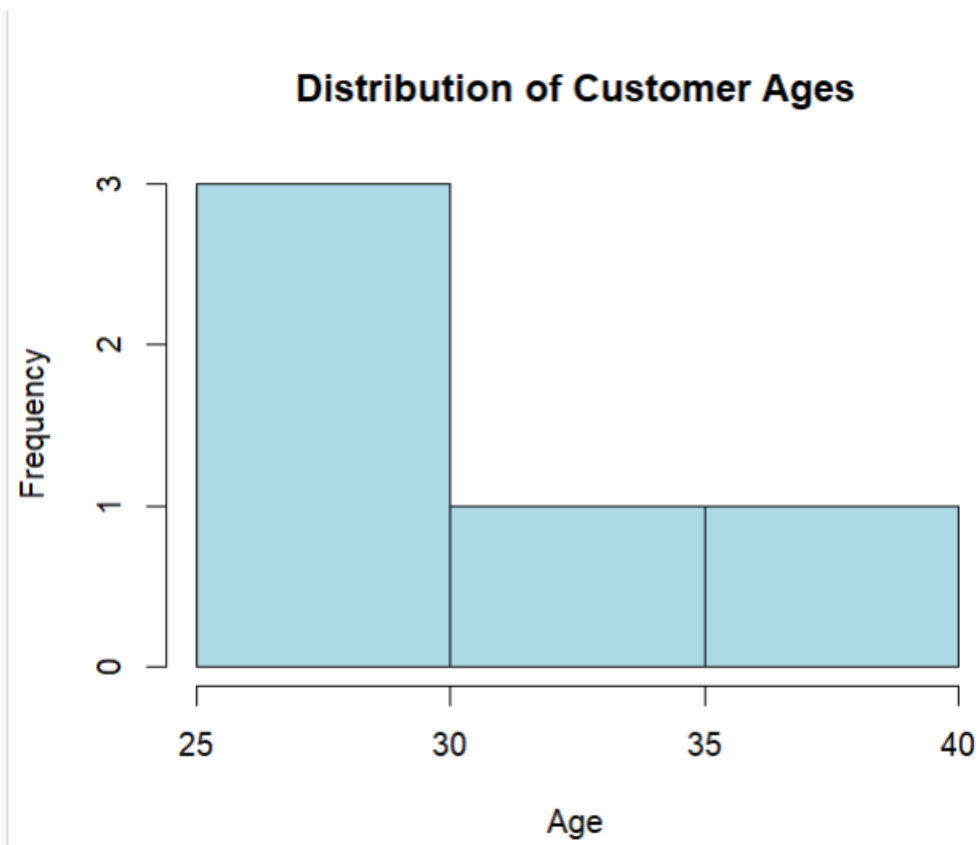
Dataset

```
customer <- data.frame(  
  CustomerID = c(1, 2, 3, 4, 5),  
  Age = c(25, 30, 35, 28, 40),  
  Satisfaction = c(4, 5, 3, 4, 5)  
)
```

Histogram

```
hist(customer$Age,  
  main = "Distribution of Customer Ages",  
  xlab = "Age",  
  ylab = "Frequency",  
  col = "lightblue",  
  border = "black",  
  breaks = 5)
```

output:

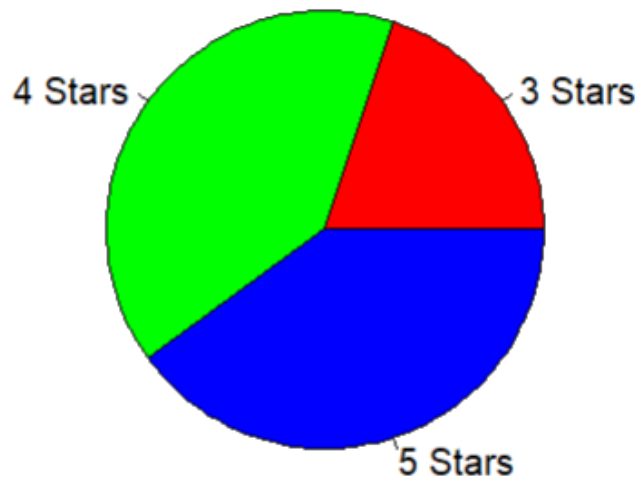


b) Generate a Pie Chart for Satisfaction Scores

R code:

```
customer <- data.frame(  
  CustomerID = c(1, 2, 3, 4, 5),  
  Age = c(25, 30, 35, 28, 40),  
  Satisfaction = c(4, 5, 3, 4, 5)  
)  
  
# Count satisfaction scores  
score_count <- table(customer$Satisfaction)  
  
# Pie Chart  
pie(score_count,  
  labels = paste(names(score_count), "Stars"),  
  col = rainbow(length(score_count)),  
  main = "Customer Satisfaction Scores")  
output:
```

Customer Satisfaction Scores



c) Create a Stacked Bar Chart by Age Group

R code:

Dataset

```
customer <- data.frame(  
  CustomerID = c(1, 2, 3, 4, 5),  
  Age = c(25, 30, 35, 28, 40),  
  Satisfaction = c(4, 5, 3, 4, 5)  
)
```

Create Age Groups

```
customer$AgeGroup <- cut(customer$Age,  
  breaks = c(20, 30, 40, 50),  
  labels = c("21-30", "31-40", "41-50"))
```

```
# Frequency Table
```

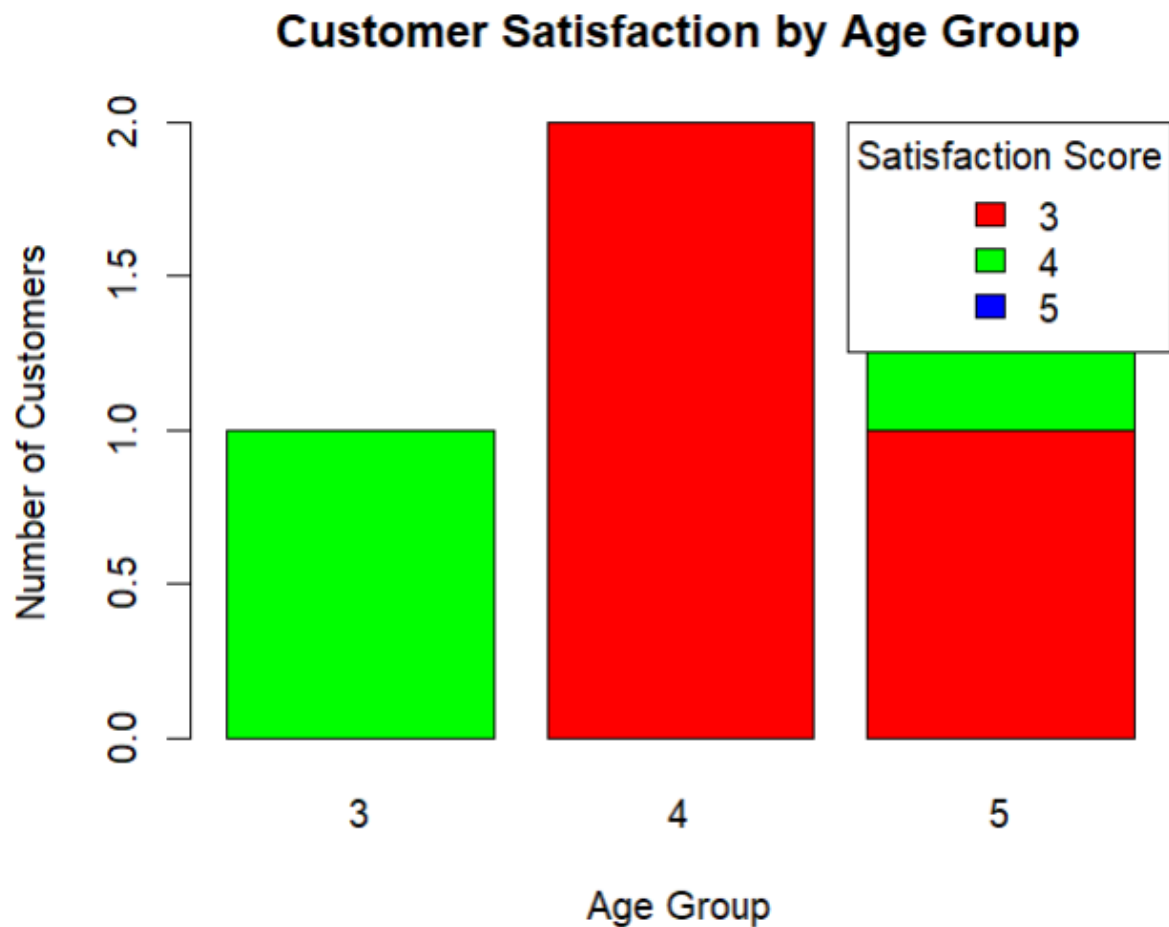
```
table_data <- table(customer$AgeGroup, customer$Satisfaction)
```

```
# Stacked Bar Chart
```

```
barplot(table_data,  
        beside = FALSE,  
        col = rainbow(ncol(table_data)),  
        main = "Customer Satisfaction by Age Group",  
        xlab = "Age Group",  
        ylab = "Number of Customers")
```

```
legend("topright",  
       legend = colnames(table_data),  
       fill = rainbow(ncol(table_data)),  
       title = "Satisfaction Score")
```

output:



d) Create a Word Cloud from Customer Feedback

R code:

```
library(tm)
```

```
library(wordcloud)
```

```
library(RColorBrewer)
```

```
# Sample Feedback
```

```
feedback <- c(
```

```
  "Excellent service and friendly staff",
```

```
  "Very satisfied with the product quality",
```

```
  "Good experience but delivery was slow",
```

```
  "Excellent customer support",
```

```
"Fast delivery and excellent quality"
)

# Create Corpus
corpus <- Corpus(VectorSource(feedback))

# Clean Text
corpus <- tm_map(corpus, content_transformer(tolower))
corpus <- tm_map(corpus, removePunctuation)
corpus <- tm_map(corpus, removeNumbers)
corpus <- tm_map(corpus, removeWords, stopwords("english"))

# Create Word Cloud
wordcloud(
  words = unlist(sapply(corpus, as.character)),
  max.words = 50,
  random.order = FALSE,
  colors = brewer.pal(8, "Dark2")
)
```

Output:

excellent

Set 3: Employee Performance Evaluation

a) Create a Line Chart for Employee Performance Trend

R code:

Dataset

```
employee <- data.frame(  
  EmployeeID = c(1, 2, 3, 4, 5),  
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),  
  YearsService = c(5, 3, 7, 4, 2),  
  Performance = c(85, 92, 78, 90, 76)  
)
```

Line Chart

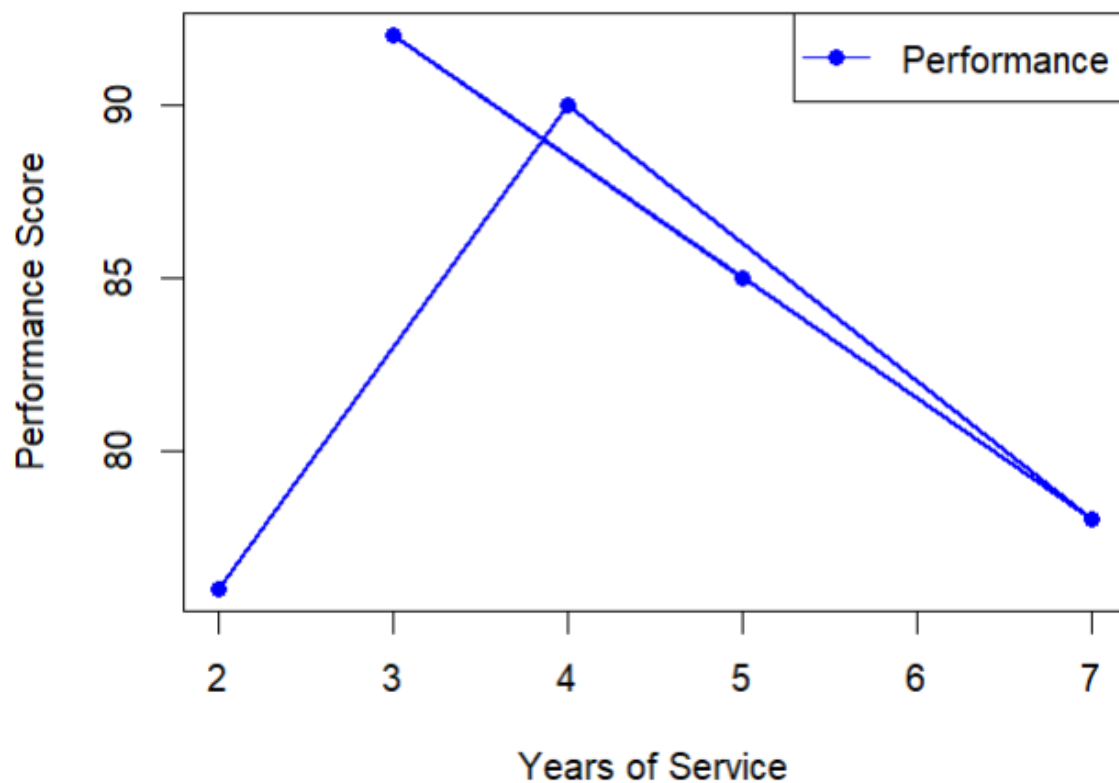
```
plot(employee$YearsService,
```

```
employee$Performance,  
type = "o",  
col = "blue",  
pch = 16,  
lwd = 2,  
xlab = "Years of Service",  
ylab = "Performance Score",  
main = "Employee Performance Trend")
```

```
legend("topright",  
      legend = "Performance",  
      col = "blue",  
      lty = 1,  
      pch = 16)
```

output:

Employee Performance Trend



b) Generate a Bar Chart Showing Employees Across Departments

R code:

R code:

Dataset

```
employee <- data.frame(  
  EmployeeID = c(1, 2, 3, 4, 5),  
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),  
  YearsService = c(5, 3, 7, 4, 2),  
  Performance = c(85, 92, 78, 90, 76)  
)
```



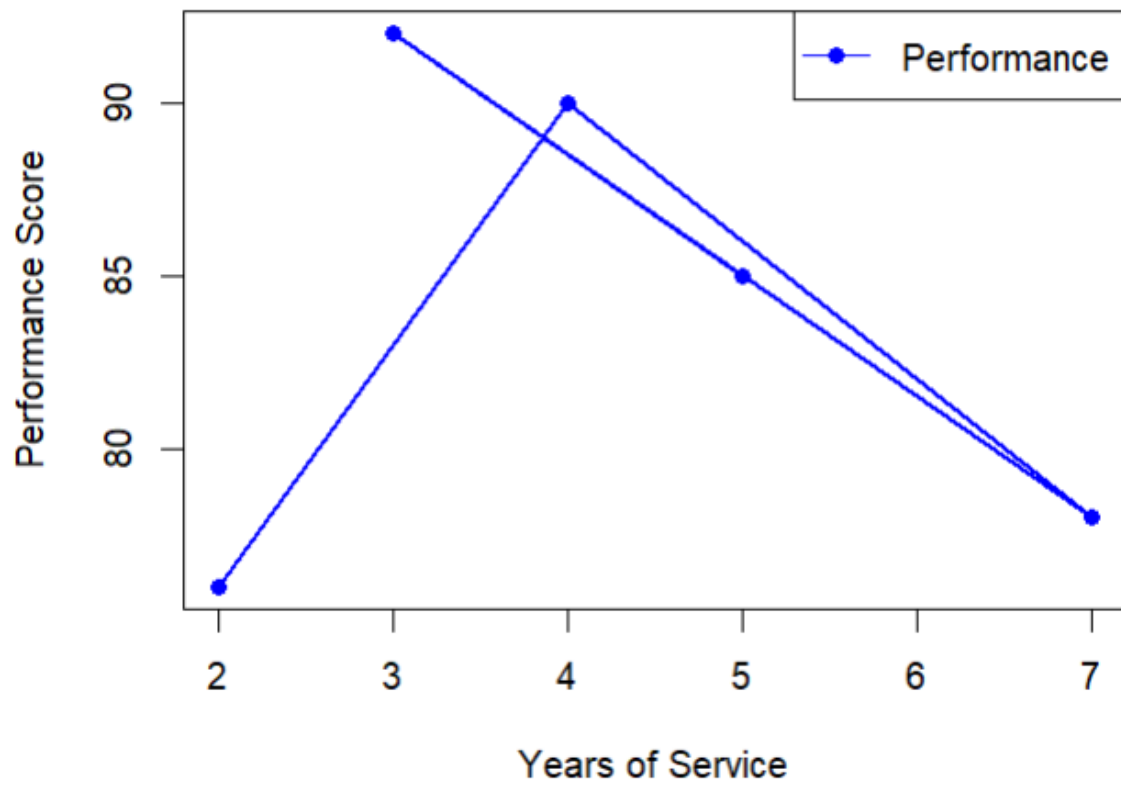
```
# Line Chart
```

```
plot(employee$YearsService,  
      employee$Performance,  
      type = "o",  
      col = "blue",  
      pch = 16,  
      lwd = 2,  
      xlab = "Years of Service",  
      ylab = "Performance Score",  
      main = "Employee Performance Trend")
```

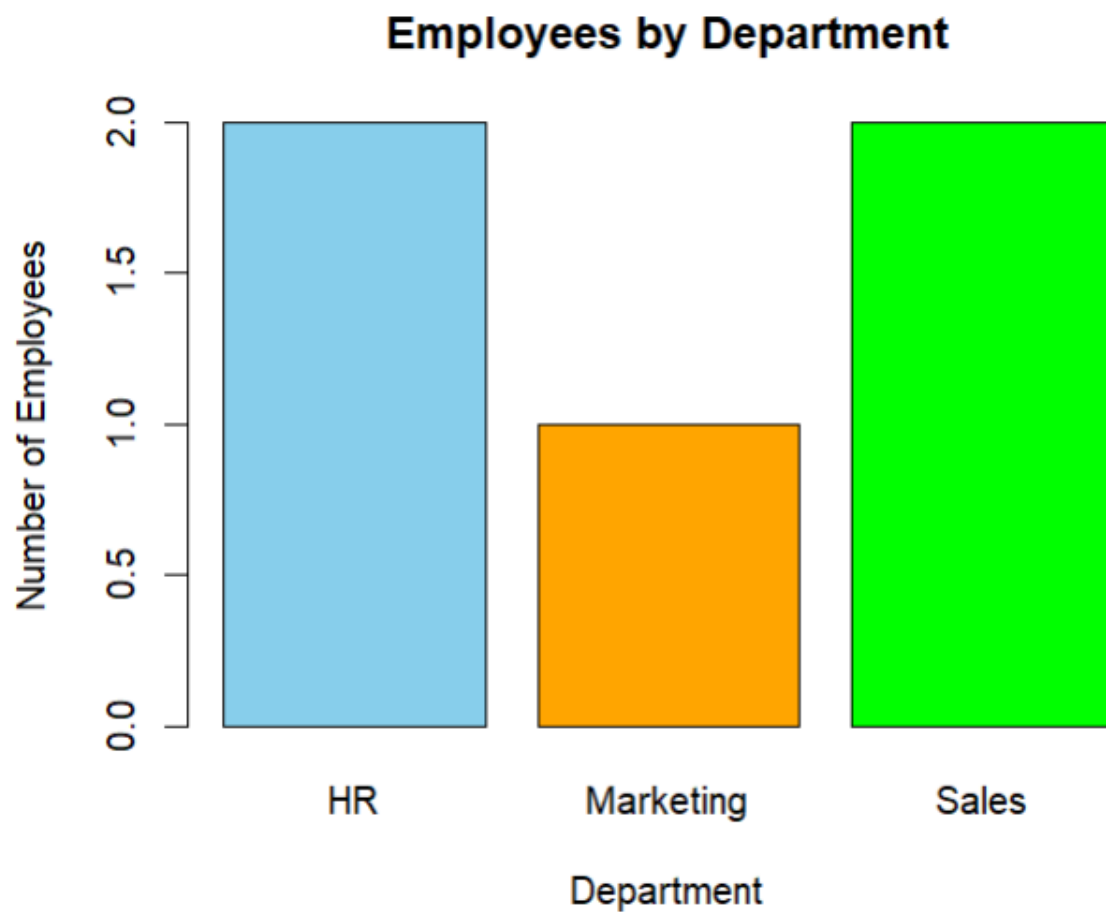
```
legend("topright",  
      legend = "Performance",  
      col = "blue",  
      lty = 1,  
      pch = 16)
```

output:

Employee Performance Trend



Output:



c) Create a Scatter Plot for Years of Service vs Performance Score

R code:

Dataset

```
employee <- data.frame(  
  EmployeeID = c(1, 2, 3, 4, 5),  
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),  
  YearsService = c(5, 3, 7, 4, 2),  
  Performance = c(85, 92, 78, 90, 76)  
)
```

Scatter Plot

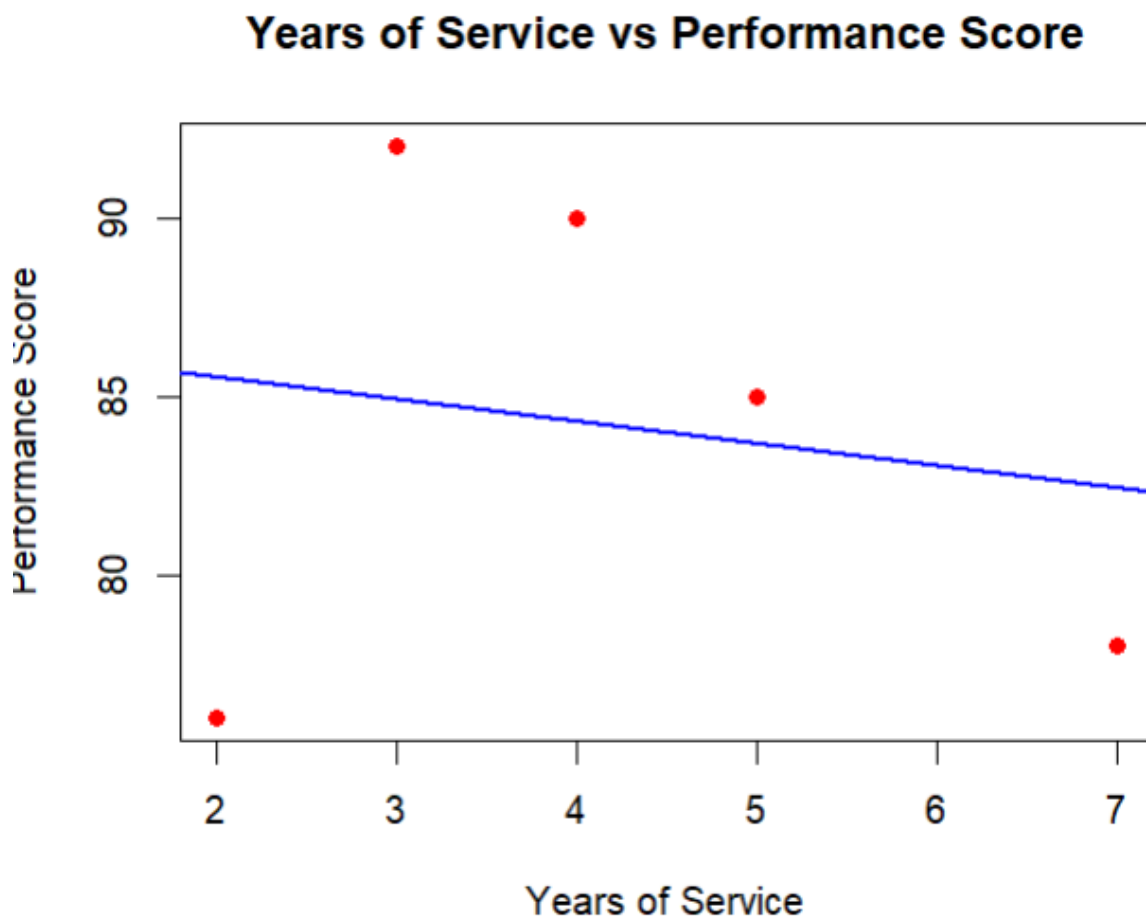
```
plot(employee$YearsService,  
      employee$Performance,
```

```
pch = 19,  
col = "red",  
xlab = "Years of Service",  
ylab = "Performance Score",  
main = "Years of Service vs Performance Score")
```

```
# Regression Line
```

```
abline(lm(Performance ~ YearsService, data = employee),  
col = "blue",  
lwd = 2)
```

output:



d) Interactive Dashboard (Using Shiny)

R code:

```
library(shiny)
```

```
employee <- data.frame(  
  EmployeeID = c(1, 2, 3, 4, 5),  
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),  
  YearsService = c(5, 3, 7, 4, 2),  
  Performance = c(85, 92, 78, 90, 76)  
)
```

```
ui <- fluidPage(  
  
  titlePanel("Employee Performance Dashboard"),  
  
  sidebarLayout(  
  
    sidebarPanel(  
  
      selectInput("chart",  
        "Choose Chart:",  
        choices = c("Line Chart",  
          "Bar Chart"))  
  
    ),  
  
    mainPanel(  
  
      plotOutput("employeePlot")
```

)

)

)

```
server <- function(input, output){
```

```
  output$employeePlot <- renderPlot({
```

```
    if(input$chart == "Line Chart"){
```

```
      plot(employee$YearsService,
            employee$Performance,
            type = "o",
            col = "blue",
            pch = 16,
            lwd = 2,
            xlab = "Years of Service",
            ylab = "Performance Score",
            main = "Employee Performance Trend")
```

```
      legend("topright",
            legend = "Performance",
            col = "blue",
            lty = 1,
            pch = 16)
```

```
}else{  
  
  dept_count <- table(employee$Department)  
  
  barplot(dept_count,  
    col = c("skyblue", "orange", "green"),  
    xlab = "Department",  
    ylab = "Number of Employees",  
    main = "Employees by Department")  
  
}  
  
})  
  
}  
  
shinyApp(ui = ui, server = server)
```

output:

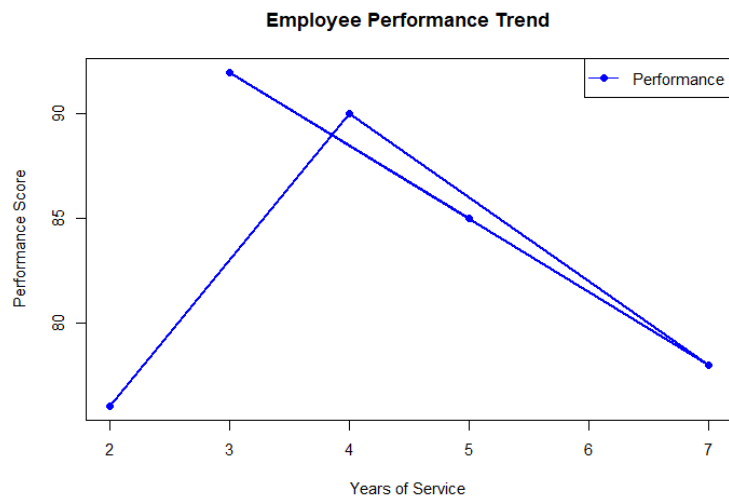
Employee Performance Dashboard



Employee Performance Dashboard

Choose Chart:

Line Chart



Set 4: Product Inventory Management

a) Create a Bar Chart for Product Inventory

R code: # Dataset

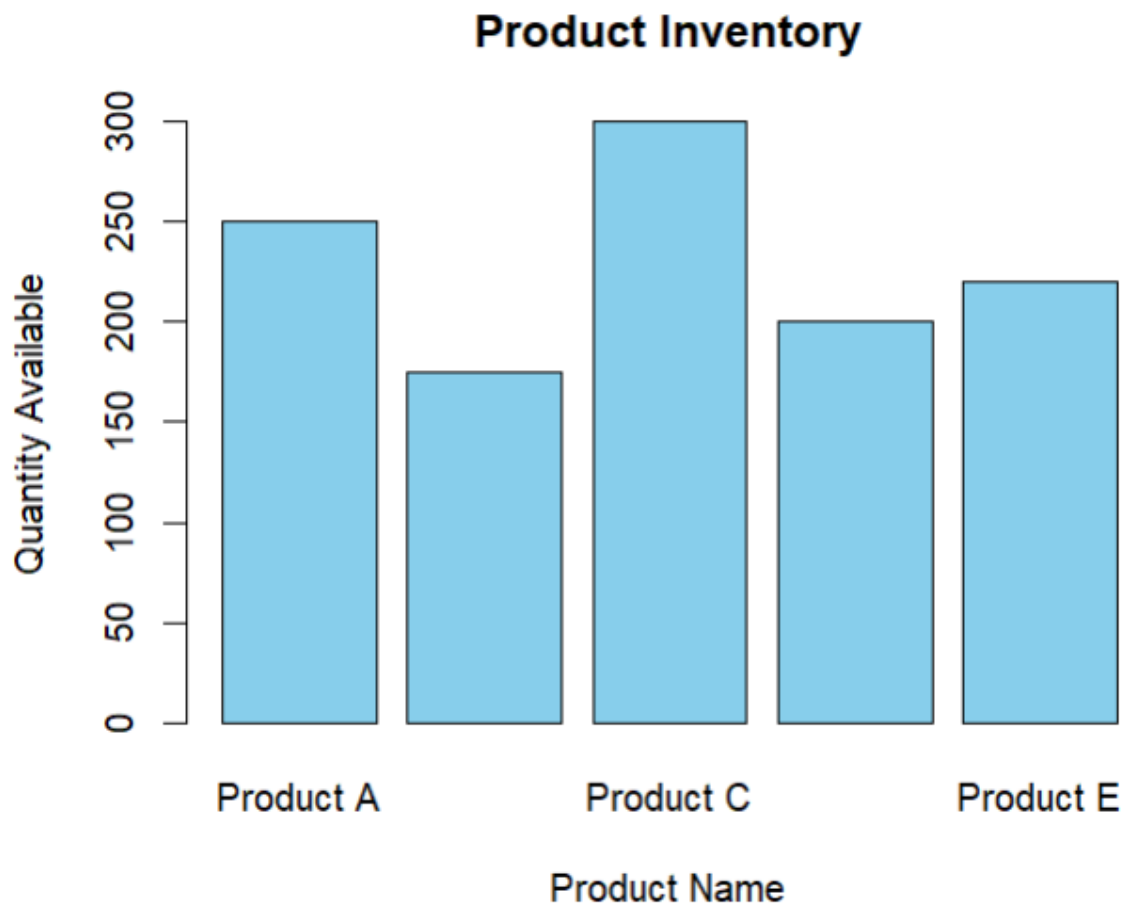
```
inventory <- data.frame(  
  ProductID = c(1, 2, 3, 4, 5),  
  ProductName = c("Product A", "Product B", "Product C", "Product D", "Product E"),  
  Quantity = c(250, 175, 300, 200, 220)  
)
```

Bar Chart

```
barplot(inventory$Quantity,  
  names.arg = inventory$ProductName,
```

```
col = "skyblue",  
xlab = "Product Name",  
ylab = "Quantity Available",  
main = "Product Inventory")
```

output:



b) Generate a Stacked Bar Chart for Product Categories

R code:

Dataset

```
inventory <- data.frame(  
  ProductName = c("Product A", "Product B", "Product C", "Product D", "Product E"),  
  Quantity = c(250, 175, 300, 200, 220),  
  Category = c("Electronics", "Electronics", "Furniture", "Furniture", "Furniture")  
)
```

```
# Create Table
```

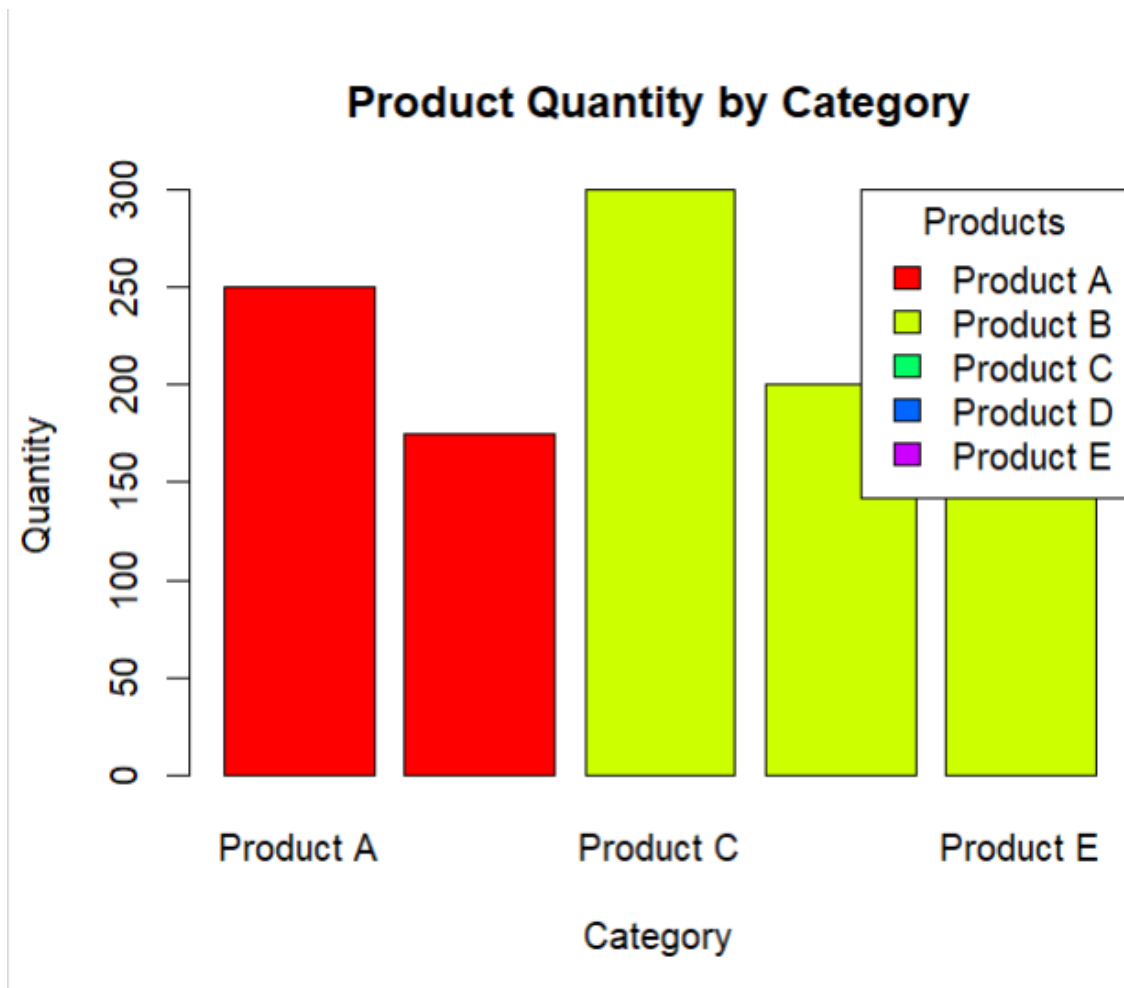
```
table_data <- xtabs(Quantity ~ Category + ProductName, data = inventory)
```

```
# Stacked Bar Chart
```

```
barplot(table_data,  
        beside = FALSE,  
        col = rainbow(ncol(table_data)),  
        main = "Product Quantity by Category",  
        xlab = "Category",  
        ylab = "Quantity")
```

```
legend("topright",  
       legend = colnames(table_data),  
       fill = rainbow(ncol(table_data)),  
       title = "Products")
```

```
output:
```



c) Create a Scatter Plot for Product Price vs Quantity

R code:

Dataset

```
inventory <- data.frame(
```

```
  ProductName = c("Product A", "Product B", "Product C", "Product D", "Product E"),
```

```
  Quantity = c(250, 175, 300, 200, 220),
```

```
  Price = c(100, 150, 80, 120, 90)
```

```
)
```

Scatter Plot

```
plot(inventory$Price,
```

```
     inventory$Quantity,
```

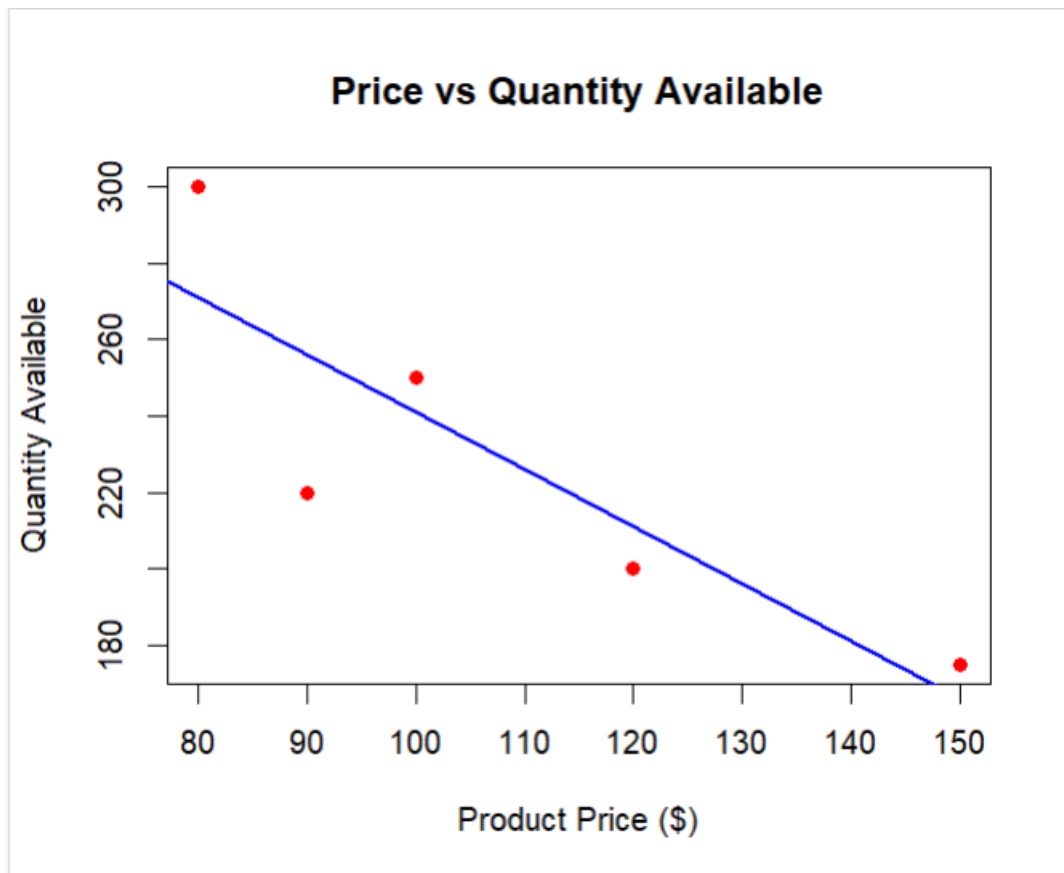
```
     pch = 19,
```

```
col = "red",  
xlab = "Product Price ($)",  
ylab = "Quantity Available",  
main = "Price vs Quantity Available")
```

```
# Regression Line
```

```
abline(lm(Quantity ~ Price, data = inventory),  
       col = "blue",  
       lwd = 2)
```

output:



d) Interactive Dashboard (Using Shiny)

R code:

```
library(shiny)
```

```
inventory <- data.frame(  
  ProductID = c(1, 2, 3, 4, 5),  
  ProductName = c("Product A", "Product B", "Product C", "Product D", "Product E"),  
  Quantity = c(250, 175, 300, 200, 220),  
  Category = c("Electronics", "Electronics", "Furniture", "Furniture", "Furniture")  
)
```

```
ui <- fluidPage(  
  
  titlePanel("Product Inventory Dashboard"),  
  
  sidebarLayout(  
  
    sidebarPanel(  
  
      selectInput("chart",  
        "Choose Chart:",  
        choices = c("Bar Chart",  
          "Stacked Bar Chart"))  
  
    ),  
  
    mainPanel(  
  
      plotOutput("inventoryPlot")  
  
    )  
  )  
)
```

)

)

```
server <- function(input, output){
```

```
  output$inventoryPlot <- renderPlot({
```

```
    if(input$chart == "Bar Chart"){
```

```
      barplot(inventory$Quantity,  
              names.arg = inventory$ProductName,  
              col = "skyblue",  
              xlab = "Product Name",  
              ylab = "Quantity Available",  
              main = "Product Inventory")
```

```
    }else{
```

```
      table_data <- xtabs(Quantity ~ Category + ProductName,  
                          data = inventory)
```

```
      barplot(table_data,  
              beside = FALSE,  
              col = rainbow(ncol(table_data)),  
              main = "Product Quantity by Category",  
              xlab = "Category",  
              ylab = "Quantity")
```

```
    legend("topright",
          legend = colnames(table_data),
          fill = rainbow(ncol(table_data)),
          title = "Products")

  }

})

}

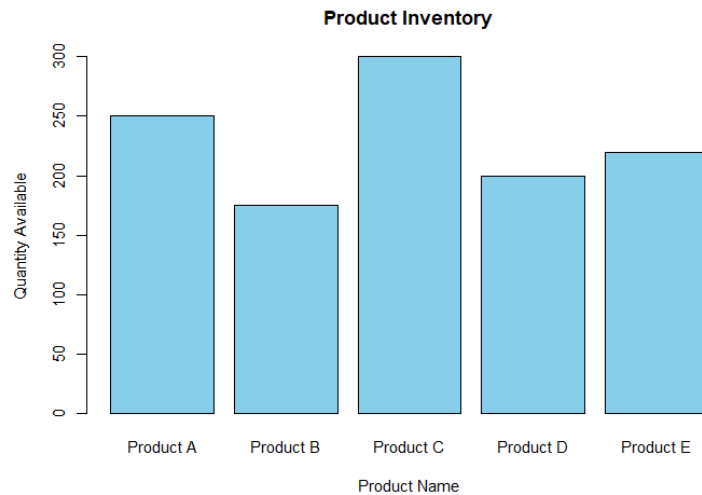
shinyApp(ui = ui, server = server)

output:
```


Product Inventory Dashboard

Choose Chart:

Bar Chart



Set 5: Website Analytics

a) Create a Line Chart for Daily Page Views

R code:

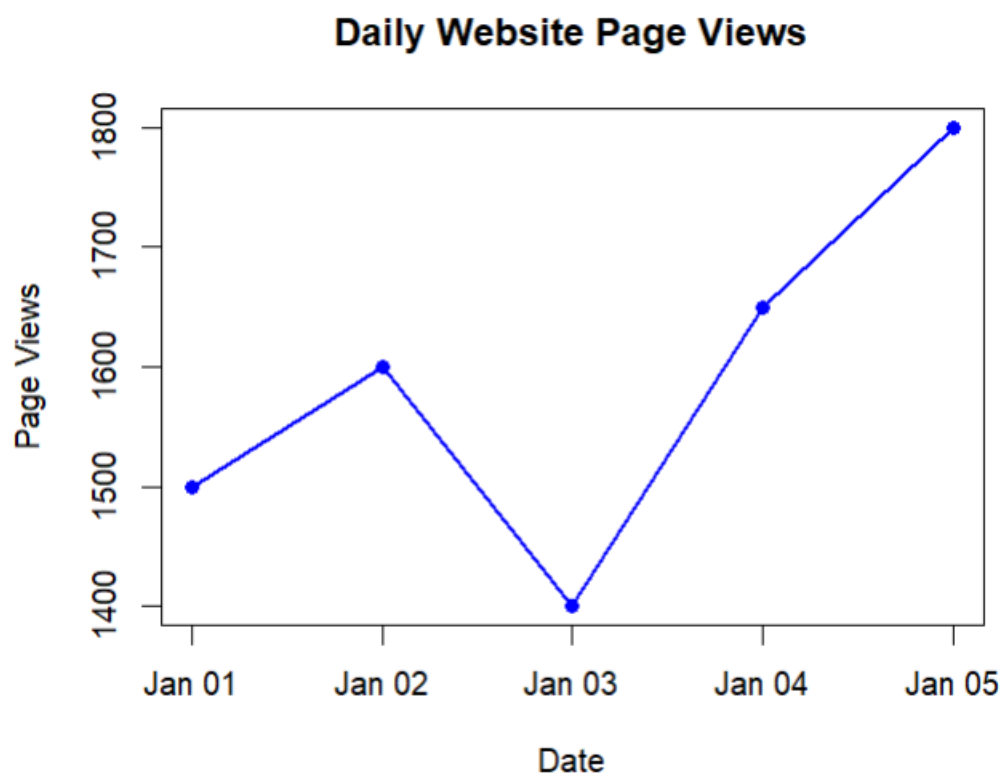
Dataset

```
traffic <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",  
    "2023-01-04", "2023-01-05")),  
  PageViews = c(1500, 1600, 1400, 1650, 1800),  
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)  
)
```

Line Chart

```
plot(traffic$Date,  
     traffic$PageViews,  
     type = "o",  
     col = "blue",  
     pch = 16,  
     lwd = 2,  
     xlab = "Date",  
     ylab = "Page Views",  
     main = "Daily Website Page Views")
```

output:



b) Generate a Bar Chart for Highest Click-through Rates

R code:

```
# Dataset
```

```
traffic <- data.frame(
```

```
Date = c("2023-01-01", "2023-01-02", "2023-01-03",  
         "2023-01-04", "2023-01-05"),  
PageViews = c(1500, 1600, 1400, 1650, 1800),  
CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)  
)
```

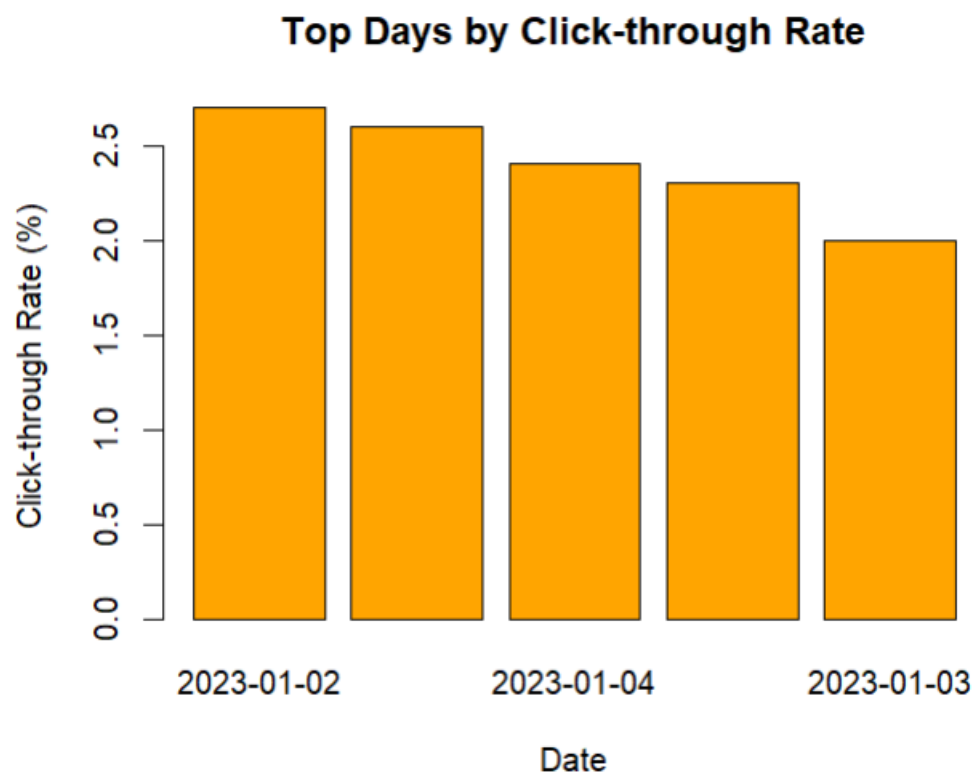
```
# Sort by CTR
```

```
traffic <- traffic[order(-traffic$CTR), ]
```

```
# Bar Chart
```

```
barplot(traffic$CTR,  
        names.arg = traffic$Date,  
        col = "orange",  
        xlab = "Date",  
        ylab = "Click-through Rate (%)",  
        main = "Top Days by Click-through Rate")
```

```
output:
```



c) Create a Stacked Area Chart for User Interactions

R code:

Sample Dataset

```
traffic <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",  
    "2023-01-04", "2023-01-05")),  
  Likes = c(200, 220, 180, 240, 260),  
  Shares = c(80, 90, 70, 85, 95),  
  Comments = c(40, 50, 35, 45, 55)  
)
```

Install package once

```
install.packages("ggplot2")
```

```
install.packages("reshape2")
```

```
library(ggplot2)
```

```
library(reshape2)
```

```
# Convert data to long format
```

```
traffic_long <- melt(traffic, id.vars = "Date")
```

```
# Stacked Area Chart
```

```
ggplot(traffic_long,
```

```
  aes(x = Date,
```

```
      y = value,
```

```
      fill = variable)) +
```

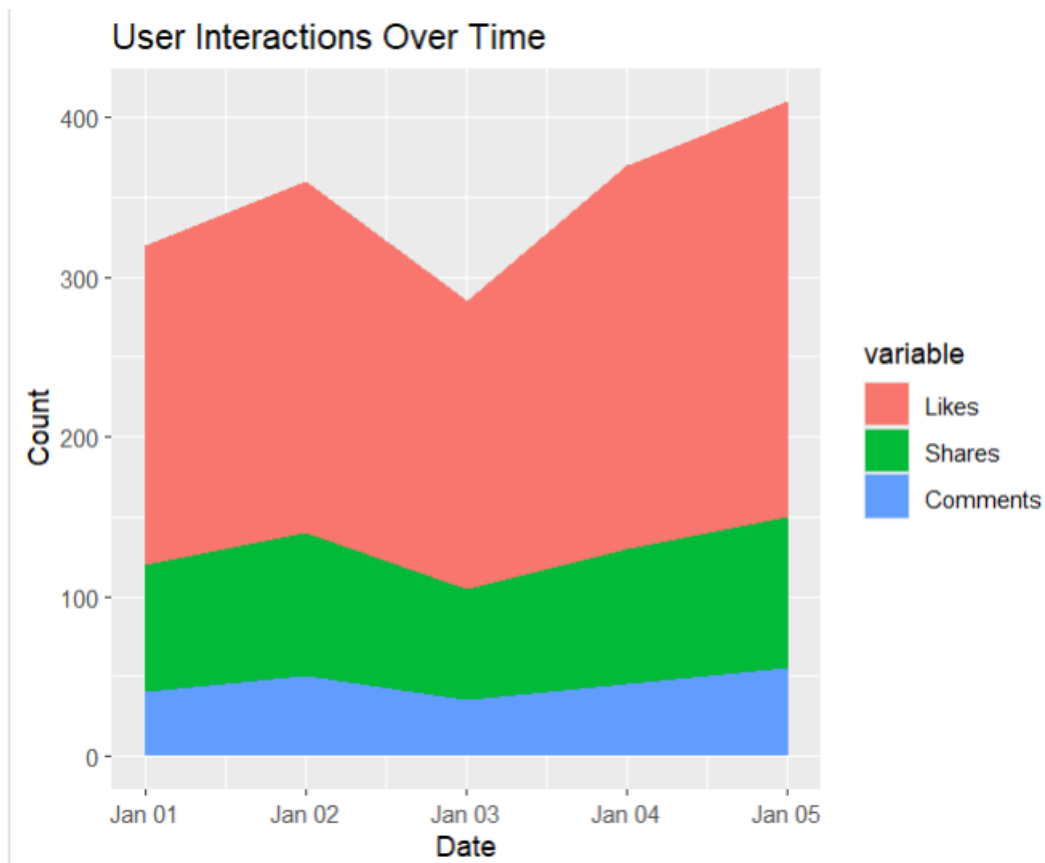
```
geom_area() +
```

```
labs(title = "User Interactions Over Time",
```

```
      x = "Date",
```

```
      y = "Count")
```

```
output:
```



d) Interactive Dashboard (Using Shiny)

R code:

```
library(shiny)
```

```
traffic <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
    "2023-01-04", "2023-01-05")),
  PageViews = c(1500, 1600, 1400, 1650, 1800),
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)
)
```

```
ui <- fluidPage(
```

```
  titlePanel("Website Analytics Dashboard"),
```

```

sidebarLayout(

  sidebarPanel(

    selectInput("chart",
               "Choose Chart:",
               choices = c("Line Chart",
                           "Bar Chart"))

  ),

  mainPanel(

    plotOutput("trafficPlot")

  )

)

)

server <- function(input, output){

  output$trafficPlot <- renderPlot({

    if(input$chart == "Line Chart"){

      plot(traffic$Date,

```

```

    traffic$PageViews,
    type = "o",
    col = "blue",
    pch = 16,
    lwd = 2,
    xlab = "Date",
    ylab = "Page Views",
    main = "Daily Website Page Views")

}else{

    temp <- traffic[order(-traffic$CTR), ]

    barplot(temp$CTR,
            names.arg = temp$Date,
            col = "orange",
            xlab = "Date",
            ylab = "Click-through Rate (%)",
            main = "Top Days by CTR")

    }

})

}

shinyApp(ui = ui, server = server)

```

output:

Website Analytics Dashboard

Choose Chart:

Line Chart

